DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2022-23
MONDAY TEST

CLASS: IX | **FULL MARKS: 40**
SUBJECT: PHYSICS | **TIME: 21.11.2022**

Instructions:
- **All questions are compulsory.**
- **This paper consists of two printed pages.**
- **Answers to sub parts of the same question must be given in one place only.**

SECTION A
(Attempt all questions.)

Question 1
Choose the correct answers to the questions from the given options. [1 × 7 = 7]

(a) A balloon filled with helium gas floats in a big closed jar which is connected to an evacuation pump. The balloon will sink when the air is pumped out. The appropriate reason for this is:
(i) Density of air in jar decreases, so the upthrust on balloon decreases
(ii) Density of air in jar increases, so the upthrust on balloon increases
(iii) Upthrust exceeds the weight of the balloon
(iv) Both (i) and (iii)

(b) A piece of wood of R.D. 0.25 floats in a pail containing oil of R.D of 0.81. The fraction of volume of the wood above the surface of the oil is
(i) 0.31 (ii) 0.69 (iii) 0.21 (iv) 0.79

(c) Which one of the following is odd?
(i) silica below −80°C (ii) silver iodide from 80°C to 141°C
(iii) water from 0°C to 4°C (iv) silica above −80°C

(d) A block of wood just floats when put in water at room temperature. What change will you observe if the water is heated?
(i) floats more than before
(ii) sinks
(iii) first sinks then float
(iv) no change

(e) If an object of mass 100 kg with a volume of $1 m^3$ is completely submerged 2 metre below the surface of water. What is the net force acting on the object? [$g = 9.8 ms^{-2}$]
(i) 9460 N (ii) 6440 N (iii) 10200 N (iv) 8820 N

(f) An object of weight 200 N is floating in a liquid. What is the magnitude of the buoyant force acting on it?
(i) zero
(ii) more than 200N
(iii) 200N
(iv) less than 200 N

(g) A body of mass 100 kg and density 500 kg/m^3 floats in water. The mass that needs to be added to allow the body to sink is
(i) 80kg (ii) 100kg (iii) 150 kg (iv) 200 kg

Question 2

(a) i) Name the specific reaction that is maintained in any nuclear power plant.
ii) How is the reaction stated above controlled?
iii) Write one limitation of using nuclear energy. [3]
(b) What is global warming? [2]
(c) Why does a block of wood held under water rise to the surface when released? [2]
(d) Distinguish between density and relative density. [2]
(e) State the principle of a solar power plant. [2]
(f) State the principle of floatation. What is the purpose of the Plimsoll line on a ship? [2]

SECTION B

Answer *all* questions

Question 3 [3+3+4]

(a) "A balloon filled with hydrogen gas rises to a certain height and then stops rising further". Explain. How are the centre of gravity of a floating object and centre of buoyancy located with respect to each other, for an object floating partially in a fluid?
(b) A sample of milk diluted with water has a density of 1060 kgm^{-3}. If pure milk has a density of 1080 $kg\ m^{-3}$ then find the percentage of water by volume in milk.
(c) Draw the graphs showing the variation
(i) of density with temperature in range of 0°C to 8°C of water
(ii) of temperature with time recorded by the lower thermometer in Hope's experiment.

Question 4 [3+3+4]

(a) State two limitations of establishing a solar panel. Mention one technological measure to minimize global warming.
(b) Write any two qualifying characteristics of a source of energy. Distinguish between renewable and non-renewable sources of energy. [any one]
(c) A solid block weighs 80.3 gf in air, 72.7 gf in water and 70.8 gf in a liquid. Find the relative density of the liquid. What is the relative density of the solid?